

TECHNICAL DATA [WL-C, WE-C]

01-50B TECHNICAL DATA [WL-C, WE-C]

ENGINE TECHNICAL DATA

[WL-C, WE-C] 01-50B-1

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DCF01500000W02

Item	Specification
Maximum cylinder head distortion	X distortion: 0.02 mm {0.0008 in} max. Y distortion: 0.05 mm {0.0020 in} max.
Maximum manifold contact surface distortion	0.05 mm {0.002 in} max.
Maximum manifold contact surface grinding	0.15 mm {0.0059 in} max.
Standard valve margin thickness	IN: 1.55—1.85 mm {0.061—0.072 in} EX: 1.80—2.10 mm {0.070—0.082 in}
Standard valve length	IN: 111.65—112.25 mm {4.394—4.413 in} EX: 111.6—112.2 mm {4.390—4.409 in}
Minimum valve length	IN: 111.50 mm {4.390 in} EX: 111.45 mm {4.388 in}
Standard valve stem diameter	IN: 5.970—5.985 mm {0.2350—0.2356 in} EX: 5.965—5.980 mm {0.2348—0.2354 in}
Minimum valve stem diameter	IN: 5.920 mm {0.2330 in} EX: 5.915 mm {0.2328 in}
Standard valve guide inner diameter	6.025—6.045 mm {0.2372—0.2379 in}
Standard valve guide height	IN: 15.0—15.5 mm {0.59—0.61 in} EX: 17.0—17.5 mm {0.67—0.69 in}
Standard valve guide height	IN: 15.0—15.5 mm {0.59—0.61 in} EX: 17.0—17.5 mm {0.67—0.68 in}
Standard valve seat contact width	1.3—1.9 mm {0.052—0.074 in}
Valve seat angle	IN: 45° EX: 45°
Standard valve recession	IN: 0.79—1.27 mm {0.039—0.050 in} EX: 0.84—1.32 mm {0.033—0.051 in}
Maximum valve recession	IN: 1.68 mm {0.066 in} EX: 1.73 mm {0.062 in}
Valve spring installation pressing force	172.9—195.6 N {15.67—17.74 kgf, 34.48—39.02 lbf}
Valve spring installation height	39.0 mm {1.53 in}
Maximum valve spring out-of-square	2.0° (1.60mm {0.062 in})
Maximum camshaft runout	0.03 mm {0.001 in} max.
Standard cam lobe height	IN: 42.067—42.167 mm {1.6561—1.6601 in} EX: 41.949—42.049 mm {1.6515—1.6554 in}
Minimum cam lobe height	IN: 41.717 mm {1.6424 in} EX: 41.599 mm {1.6377 in}
Standard cam journal diameter	No.1: 31.940—31.965 mm {1.2575—1.2582 in} No.2—No.4: 25.910—25.935 mm {1.0201—1.0210 in} No.5: 25.940—25.965 mm {1.0212—1.0222 in}
Minimum cam journal diameter	No.1: 31.890 mm {1.2555 in} No.2—No.4: 25.860 mm {1.0181 in} No.5: 25.890 mm {1.0193 in}
Standard camshaft clearance	No.1: 0.035—0.081 mm {0.0014—0.0031 in} No.2—4: 0.065—0.111 mm {0.0026—0.0043 in} No.5: 0.056—0.081 mm {0.0014—0.0031 in}
Maximum camshaft clearance	No.1: 0.12 mm {0.0047 in} No.2—4: 0.15 mm {0.0059 in} No.5: 0.16 mm {0.0063 in}
Standard camshaft end play	0.030—0.160 mm {0.0012—0.0062 in}
Maximum camshaft end play	0.20 mm {0.0078 in}
Maximum cylinder block distortion	X direction: 0.02 mm {0.0008 in} Y direction: 0.05 mm {0.002 in}
Cylinder bore size [WL-C]	Standard: 93.000—93.022 mm {3.6615—3.6622 in} 0.25 {0.01} oversize: 93.250—93.272 mm {3.6713—3.6721 in} 0.50 {0.02} oversize: 93.500—93.522 mm {3.6811—3.6819 in}

01-50B-1

TECHNICAL DATA [WL-C, WE-C]

Item	Specification
Cylinder bore size [WE-C]	Standard: 96.000—96.022 mm {3.6615—3.6622 in} 0.25 {0.01} oversize: 96.250—96.272 mm {3.7893—3.7902 in} 0.50 {0.02} oversize: 96.500—96.522 mm {3.7992—3.8000 in}
Cylinder bore wear limit	0.15 mm {0.0059 in}
Flywheel maximum runout	1.5 mm {0.059 in}
Oil jet air pressure	137.6—196.4 kPa {1.4—2.0 kgf/cm ² , 20—28 psi}
Piston diameter [WL-C]	Standard: 92.918—92.944 mm {3.6582—3.6592 in} 0.25 {0.010} oversize: 93.153—93.179 mm {3.6675—3.6684 in} 0.50 {0.020} oversize: 93.403—93.429 mm {3.6773—3.6782 in}
Piston diameter [WE-C]	Standard: 95.918—95.944 mm {3.7763—3.7773 in} 0.25 {0.010} oversize: 96.153—96.179 mm {3.7856—3.7865 in} 0.50 {0.020} oversize: 96.403—96.429 mm {3.7954—3.7964 in}
Standard piston clearance	0.071—0.089 mm {0.0015—0.0022 in}
Maximum piston clearance	0.15 mm {0.0059 in}
Standard piston ring clearance [WL-C]	Top: 0.06—0.10 mm {0.0024—0.0039 in} Second: 0.06—0.08 mm {0.0024—0.0031 in} Oil: 0.02—0.06 mm {0.0012—0.0023 in}
Standard piston ring clearance [WE-C]	Top: 0.06—0.10 mm {0.0024—0.0039 in} Second: 0.04—0.08 mm {0.0016—0.0031 in} Oil: 0.02—0.06 mm {0.0008—0.0023 in}
Maximum piston ring clearance	0.15 mm {0.0059 in}
Standard piston ring end gap [WL-C]	Top: 0.22—0.32 mm {0.0087—0.0125 in} Second: 0.49—0.64 mm {0.0193—0.0251 in} Oil: 0.22—0.52 mm {0.0087—0.0204 in}
Standard piston ring end gap [WE-C]	Top: 0.23—0.33 mm {0.0091—0.0129 in} Second: 0.50—0.65 mm {0.0197—0.0255 in} Oil: 0.22—0.52 mm {0.0087—0.0204 in}
Maximum piston ring end gap	1.0 mm {0.039 in}
Standard piston pin bore diameter	33.997—34.007 mm {1.3384—1.3388 in}
Standard connecting rod small end inner diameter	34.012—34.033 mm {1.3391—1.3398 in}
Standard piston pin diameter	33.994—34.000 mm {1.3384—1.3385 in}
Standard piston pin-to-piston pin bore clearance	−0.003—0.013 mm {−0.0001—0.0005 in}
Standard connecting rod small end-to-piston pin clearance	0.012—0.039 mm {0.00048—0.0015 in}
Maximum crankshaft runout	0.05 mm {0.002 in}
Main journal diameter [No.1,2,4,5]	Standard: 66.937—66.955 mm {2.6354—2.6360 in} 0.25 {0.010} undersize: 66.687—66.705 mm {2.6255—2.6261 in} 0.50 {0.020} undersize: 66.437—66.455 mm {2.6157—2.6163 in} 0.75 {0.030} undersize: 66.187—66.205 mm {2.6058—2.6064 in}
Main journal diameter [No.3]	Standard: 66.920—66.938 mm {2.6347—2.6353 in} 0.25 {0.010} undersize: 66.670—66.688 mm {2.6367—2.6373 in} 0.50 {0.020} undersize: 66.420—66.438 mm {2.6150—2.6156 in} 0.75 {0.030} undersize: 66.170—66.188 mm {2.6052—2.6058 in}
Main journal wear limit	0.05 mm {0.002 in}
Main journal out-of-round	0.03 mm {0.001 in}
Crank pin diameter [WL-C]	Standard: 54.940—54.955 mm {2.1630—2.1635 in} 0.25 {0.010} undersize: 54.690—54.705 mm {2.1532—2.1537 in} 0.50 {0.020} undersize: 54.440—54.455 mm {2.1434—2.1438 in} 0.75 {0.030} undersize: 54.190—54.205 mm {2.1335—2.1340 in}
Crank pin diameter [WE-C]	Standard: 56.940—56.955 mm {2.2417—2.2423 in} 0.25 {0.010} undersize: 56.690—56.705 mm {2.2318—2.2324 in} 0.50 {0.020} undersize: 56.440—56.455 mm {2.2220—2.2226 in} 0.75 {0.030} undersize: 56.190—56.205 mm {2.2122—2.2128 in}
Crank pin wear limit	0.05 mm {0.0020 in}
Crank pin out-of-round	0.03 mm {0.0012 in}
Standard main journal clearance	No.1, 2, 4, 5: 0.027—0.045 mm {0.0010—0.0017 in} No.3: 0.044—0.062 mm {0.0017—0.0024 in}
Maximum main journal clearance	0.08 mm {0.003 in}

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Item	Specification
Main bearing thickness	Standard: 2.006—2.021 mm {0.0789—0.0794 in} 0.25 {0.010} undersize: 2.124—2.134 mm {0.0836—0.0838 in} 0.50 {0.020} undersize: 2.249—2.259 mm {0.0885—0.0888 in} 0.75 {0.030} undersize: 2.374—2.384 mm {0.0934—0.0937 in}
Standard crankshaft end play	0.040—0.282 mm {0.0016—0.0111 in}
Maximum crankshaft end play	0.3 mm {0.01 in}
Thrust bearing thickness	Standard: 2.455—2.505 mm {0.0967—0.986 in} 0.35 {0.010} oversize: 2.630—2.680 mm {0.1036—0.1055 in}
Connecting rod bending	0.075 mm {0.0030 in} max./50 mm {2.0 in}
Connecting rod distortion	0.18 mm {0.0070 in} max./50 mm {1.968 in}
Connecting rod center-to-center distance [WL-C]	162.96—163.04 mm {5.983—5.986 in}
Connecting rod center-to-center distance [WE-C]	157.96—158.04 mm {5.983—5.986 in}
Standard connecting rod oil clearance	0.025—0.052 mm {0.0009—0.0020 in}
Maximum connecting rod oil clearance	0.08 mm {0.003 in}
Connecting rod bearing thickness	Standard: 1.507—1.516 mm {0.0592—0.0595 in} 0.25 {0.010} undersize: 1.624—1.634 mm {0.0638—0.0642 in} 0.50 {0.020} undersize: 1.749—1.759 mm {0.0687—0.0691 in} 0.75 {0.030} undersize: 1.874—1.884 mm {0.0737—0.0740 in}
Standard connecting rod side clearance	0.110—0.262 mm {0.0043—0.0103 in}
Maximum connecting rod side clearance	0.35 mm {0.014 in}
Balance shaft Standard end play	0.04—0.16 mm {0.002—0.006 in}
Standard diameter	Front: 41.945—41.960 mm {1.6514—1.6519 in} Center: 39.945—39.960 mm {1.5727—1.5732 in} Rear: 37.975—37.990 mm {1.4951—1.4956 in}
Standard clearance	Front, rear: 0.050—0.115 mm {0.0020—0.0045 in} Center: 0.080—0.145 mm {0.0032—0.0057 in}
Cylinder head bolt length	Bolt head mark W Standard length: 101.2—101.8 mm {3.985—4.007 in} Maximum length: 102.5 mm {4.035 in} Bolt head mark N Standard length: 113.2—113.8 mm {4.457—4.480 in} Maximum length: 114.5 mm {4.508 in} Bolt head mark I Standard length: 149.0—150.0 mm {5.866—5.905 in} Maximum length: 150.5 mm {5.925 in}
Main bearing cap bolt length	Standard length: 84.7—85.3 mm {3.34—3.35 in} Maximum length: 86.0 mm {3.39 in}
Connecting rod cap bolt length	Standard length: 55.45—56.05 mm {2.19—2.20 in} Maximum length: 56.75 mm {2.23 in}
Valve clearance [Engine cold]	IN: 0.10—0.16 mm {0.0040—0.0062 in} EX: 0.17—0.23 mm {0.0067—0.0090 in}
Valve clearance [Engine cold]	IN: 0.10—0.16 mm {0.0040—0.0062 in} EX: 0.17—0.23 mm {0.0067—0.0090 in}
Standard oil pump tip clearance	0.10—0.19 mm {0.0040—0.0074 in}
Maximum oil pump tip clearance	0.20 mm {0.008 in}
Standard oil pump side clearance	0.04—0.09 mm {0.0016—0.0035 in}
Maximum oil pump side clearance	0.15 mm {0.0059 in}
Standard plunger spring length	43.8 mm {1.72 in}
Front oil seal press-in amount	0.0—0.40 mm {0.0—0.015 in}
Fuel injection pump oil seal press-in amount	0.0—0.40 mm {0.0—0.015 in}
Camshaft oil seal press-in amount	0.5—1.0 mm {0.020—0.039 in}

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